PORTFOLIO DOCUMENT · MULTI-AGENT AI SYSTEM

Sovereign Advisor

AI-Powered Financial Advisory

A fully orchestrated, multi-agent AI system that conducts structured financial fact-finds, matches clients to suitable products via RAG, and generates personalised adviser briefings using GPT-4o — deployed in Microsoft Teams.

Microsoft Copilot Studio | Azure AI Foundry | FastMCP · Python | GPT-4o | SharePoint RAG | Microsoft Teams

github.com/gayatrianne/sovereign-advisor

01 / THE CHALLENGE

Why traditional advisory tools fall short

Financial advisers spend significant time on data gathering, product research and report writing. An orchestrated AI system handles this end-to-end, leaving advisers free to focus on client relationships and complex judgements.

THE PROBLEM

- Manual fact-finds are time-consuming and inconsistent
- Product matching requires cross-referencing multiple sources
- Suitability reports take hours to draft and format
- No single platform orchestrates the full advisory workflow

THE SOLUTION

- Conversational AI fact-find in Teams, one question at a time
- RAG product matching over SharePoint knowledge base
- Live financial calculations via 8 MCP tools
- GPT-4o generates a structured adviser briefing in seconds

3	8	7	1
AI AGENTS	MCP TOOLS	KB DOCUMENTS	TEAMS DEPLOY

Conversational Fact-Find | RAG Product Matching | Live Calculations | Structured Briefing | Summary Confirmation | Hold Message

github.com/gayatrianne/sovereign-advisor

02 / HOW IT WORKS

End-to-end advisory in six steps

01

CONVERSATIONAL FACT-FIND

The Sovereign Advisor orchestrator greets the client in Microsoft Teams and conducts a structured fact-find, asking one question at a time in natural language. No forms, no switching apps. Questions follow KB007 and the Required-Fields-by-Persona rules.

02

SUMMARY CONFIRMATION

Before any analysis begins, the orchestrator presents a structured summary of everything collected. The client can review and correct any detail, ensuring accuracy before the adviser briefing is generated.

03

RAG PRODUCT MATCHING

The orchestrator invokes the Product Matcher child agent, which runs RAG over the SharePoint knowledge base (KB001–KB006) and returns a structured list of matched products. No user-facing output at this stage.

04

LIVE FINANCIAL CALCULATIONS

The Sovereign-Suitability-Engine (Azure AI Foundry) receives client data and matched products. It calls the FastMCP server's 8 financial calculation tools on demand — pension gap, mortgage affordability, savings gap, JISA projection and more — using live figures.

05

STRUCTURED ADVISER BRIEFING

GPT-4o generates a professional adviser briefing — personalised recommendations backed by live calculations, structured by product area, free of hallucinated citations. Returned to the orchestrator via Activity Protocol v2 (A2A).

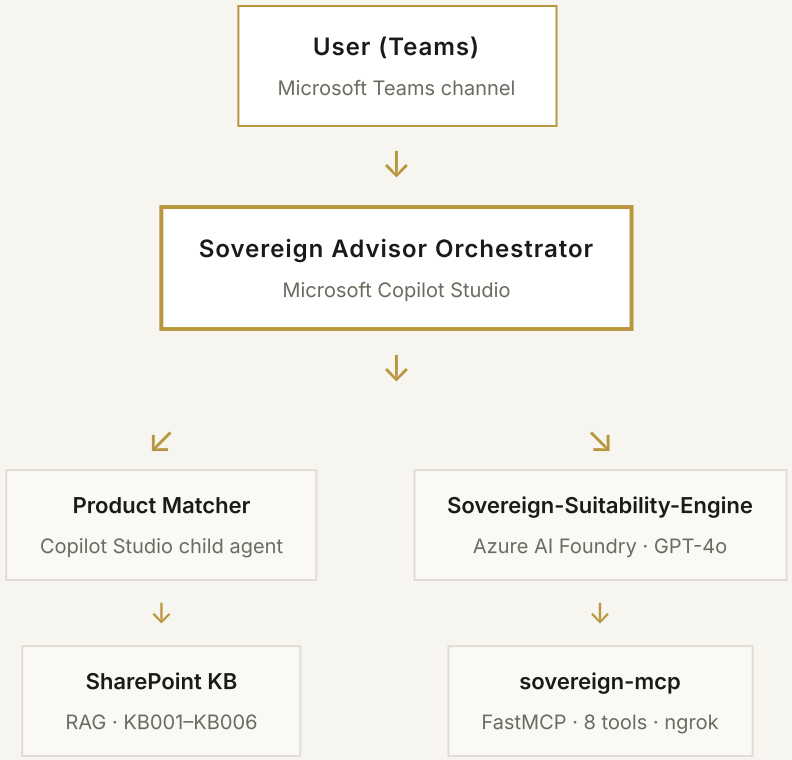
06

DELIVERY IN TEAMS

The complete briefing is delivered back to the client in the Microsoft Teams conversation. The entire flow — from first question to final briefing — runs inside Teams with no external navigation required.

03 / ARCHITECTURE

Multi-agent orchestration diagram



Orchestrator → Child Agent: invoke for KB RAG only; returns matched products, no user output

Foundry Agent → MCP Server: SSE/Streamable-HTTP via ngrok; flat key-value parameters

Orchestrator → Foundry Agent: Activity Protocol v2 (A2A); structured key-value handoff

Two-hop design: eliminates extra LLM chain, resolves >15 min timeout in three-hop design

github.com/gayatrianne/sovereign-advisor

Sovereign Advisor

04 / COPILOT STUDIO CONFIGURATION

Orchestrator and agents in Copilot Studio

Sovereign Advisor agent overview — name, description, and instructions

Agents tab showing Product Matcher (child) and Sovereign-Suitability-Engine (connected)

Knowledge tab showing SharePoint SovereignAdvisor knowledge source

Product Matcher child agent — overview with system instructions

Product Matcher knowledge tab — SharePoint KB documents

05 / AZURE AI FOUNDRY & MCP SERVER

Suitability engine and financial calculation tools

Azure AI Foundry — Sovereign-Suitability-Engine agent details page

Foundry tools tab showing sovereign-mcp MCP server connected

VS Code terminal — python main.py running with all 8 MCP tools registered

MCP FINANCIAL CALCULATION TOOLS

debt_cost Monthly interest cost and total debt burden	**transfer_fee_vs_interest** Balance transfer fee vs. interest saving comparison	**savings_gap** Gap between current trajectory and savings target
emergency_fund_target Recommended emergency fund size by persona	**pension_gap** Projected pension shortfall at retirement	**drawdown_sustainability** Years a pension pot will last in drawdown
mortgage_affordability Maximum borrowing and monthly repayment estimate	**jisa_projection** Junior ISA growth projection to age 18	

06 / LIVE IN TEAMS

The complete advisory conversation

From opening greeting to personalised adviser briefing — every step runs inside a single Microsoft Teams conversation.



Opening adviser message delivered in Microsoft Teams



Sovereign Advisor asking follow-up questions one at a time



Summary confirmation message — client reviews collected data

06 / ADVISER BRIEFING OUTPUT

GPT-4o suitability briefing delivered in Teams

The Sovereign-Suitability-Engine returns a structured, personalised briefing backed by live MCP calculations. Each section addresses a specific product area with actionable recommendations.



Adviser briefing — Emergency Fund and Debt Management sections



Adviser briefing — Pension Planning and Child Savings sections

07 / COMPONENTS & KEY DECISIONS

Technical stack and design choices

COMPONENT	TECHNOLOGY	ROLE
Orchestrator	Microsoft Copilot Studio	Conducts fact-find, holds client, routes to child & connected agents
Child Agent	Copilot Studio	RAG over SharePoint KB; returns matched products only
Suitability Engine	Azure AI Foundry · GPT-4o	Receives client data + matched products; calls MCP tools; returns briefing
Knowledge Base	SharePoint (7 documents)	Product KB (KB001–006), fact-find questions (KB007), persona fields
Calculation Tools	FastMCP · Python	8 financial tools with Pydantic validation; SSE/Streamable-HTTP transport
Tunnel	ngrok	Exposes local MCP server to Azure AI Foundry over HTTPS
Deployment	Microsoft Teams	Published via Copilot Studio → Channels → Microsoft Teams

DESIGN DECISIONS

FLAT MCP PARAMETERS

Azure AI Foundry sends tool calls as flat key-value pairs. All 8 MCP tools accept flat parameters and construct Pydantic models internally, mapping field names as needed (e.g. *pot_gbp* → *current_pot_gbp*).

TWO-HOP ARCHITECTURE

An Orchestrator → Child → Connected Agent chain caused timeouts exceeding 15 minutes. Restructuring to two hops — child handles RAG only, Foundry called

directly — eliminated one LLM chain and resolved the timeout.

STRUCTURED KEY-VALUE HANDOFF

Copilot Studio wraps connected agent calls in natural language. Using structured key-value text (not JSON) for the handoff ensures reliable parsing by the Foundry agent without format ambiguity.

ONE-QUESTION UX

The orchestrator asks exactly one question per turn throughout the fact-find. This mirrors a real advisory conversation and reduces cognitive load — clients answer naturally rather than completing a form.

08 / ABOUT THE AUTHOR

Gayatri Anne

AI & Cloud Architect | 18+ Years Enterprise IT

I build AI-powered automation systems that eliminate manual work from business processes, combining agentic workflows, large language models and cloud integration to deliver production-ready solutions.

CERTIFICATIONS

- Azure Solutions Architect Expert
- Azure AI Engineer Associate
- Azure Administrator
- Azure AI Fundamentals
- Azure Fundamentals
- TOGAF Foundation
- Python (PCAP)
- Microsoft Certified: AI Agent Builder Associate